Liver Injury — R Factor (DILI pattern) & King's College Criteria

drug-induced liver injury · acute liver failure · ward / casualty

R = ALT÷ALP (×ULN)

R FACTOR

R +: **(ALT ÷ ALT ULN) ÷ (ALP ÷ ALP ULN)**

PATTERN

Hepatocellular ALT ≥3 × ULN and R ≥5 — INH, PZA, NVP, EFV, viral hepatitis, IRIS +

Mixed R >2 to <5

Cholestatic ALP ≥2 × ULN and R ≤2 — rifampicin, amoxicillin-clavulanate, cephalosporins, sulfonylureas +

SA STG — ALT ON ART

- ALT <2.5 × ULN: continue; repeat in 2 weeks +
- ALT 2.5–5 × ULN: continue; repeat in 1 week +
- ALT >5 × ULN: stop ART (earlier if symptoms or jaundice); hepatitis screen, LFT, INR +

KING'S COLLEGE CRITERIA — REFER FOR TRANSPLANT

Paracetamol-induced

- ▸ Arterial pH <7.30 after fluid resuscitation, OR
- ▸ ALL of: INR >6.5 + creatinine >300 µmol/L + grade III–IV encephalopathy
- ▸ Modified: lactate >3.5 mmol/L at 4 h or >3.0 after fluids

Non-paracetamol

- ▸ INR >6.5, OR any 3 of:
- ▸ Age <10 or >40 y · non-A non-B hepatitis or drug-induced
- ▸ Jaundice → encephalopathy >7 days · INR >3.5 · bilirubin >300 µmol/L

SA STG — PARACETAMOL

- Children — refer severe hepatotoxicity: INR >2 at 24 h or >3 any time, pH <7.3, HCO_3 <18 or lactate >3, hypotension despite fluids, encephalopathy, creatinine >200 +
- Adults with liver failure → refer for possible transplant +
- NAC IV 200 mg/kg over 4 h, then 100 mg/kg over 16 h +

TBPAED Management of TB in Children & Adolescents (DS-TB), NDoH Sept 2024 — pp 68 · **HOSP** STG & EML Hospital Level Adults, 6th ed. 2024 (updated Mar 2026) — pp 10.16, 19.18, 19.19 · **PAED** STG & EML Paediatric Hospital Level, 2025-2029 cycle (updated Mar 2026) — pp 18.20 · **DILI** Chalasani NP et al. ACG DILI guideline. Am J Gastroenterol 2021;116:878-98 · **KINGS** O'Grady JG et al. Gastroenterology 1989;97:439-45; Bernal W et al. Lancet 2002;359:558-63

+ = SA STG / national guideline (pages listed above). Unmarked content: international source named above. ‡ = STG misprint or sources differ (see app). ANDH · liver · updated 2026-10-14